Guidelines

N-Acetylcystine (NAC) IV Infusion for Paracetamol Poisoning

Scope

Site	Service/Department/Unit	Disciplines
SCGH	Critical Care Areas, Medical & Surgical Wards	Medical, Nursing

Introduction

This guideline provides guidance to all Sir Charles Gairdner Hospital (SCGH) staff involved in the prescription, administration, supply and management of patients treated with / administered N-Acetylcysteine. This document supports and complies with related legislation, policy and other related documents here.

1.0 General Information

Paracetamol overdose can cause hepatotoxicity and potentially fatal fulminant liver failure. Signs of paracetamol poisoning include vomiting, right upper quadrant tenderness and mental state changes, but potentially life-threatening poisoning may be clinically silent for the first 12 – 24 hours.

NAC (N-acetylcysteine) effectively reduces mortality when started within 8 hours of paracetamol overdose. Prognosis is also improved when therapy is initiated later in the clinical course of hepatotoxicity. If hepatic failure develops, intensive supportive care and consultation with hepatology is required, including consideration of liver transplantation. Hepatocytes will regenerate and return to normal function without long term sequelae if the patient recovers from the initial hepatic insult.[1]

A 2 bag NAC protocol significantly reduces the rate of adverse effects during administration[2]. The loading dose is given over 4 hours thus reducing the occurrence of anaphylactoid reactions which can occur after rapid NAC loading.[3]

2.0 Presentation

- Acetylcysteine 2g in 10ml (200mg/ml) concentrated injection ampoule for IV infusion only. Must be diluted before use.

3.0 Indication

- Single ingestion of immediate release paracetamol >10g or >200mg/kg (whichever is less) AND <500mg/kg, presenting to hospital within 8 hours, where a timed paracetamol level indicates the patient is at risk of hepatotoxicity (i.e. above the treatment nomogram line).
- Modifications to this standard management are required in staggered ingestions; repeated supra-therapeutic ingestion; massive ingestion >500mg/kg; slow-release

preparation ingestion; if the plotted level is twice the treatment threshold of the nomogram; or if NAC is started greater than 8 hours post ingestion[1,4]. Further Toxicology or Hepatology advice should be sought in these scenarios.

4.0 Contraindications

- When a timed paracetamol level indicates low risk for hepatotoxicity (i.e. is below the treatment nomogram in paracetamol overdose).

5.0 Precautions

- Observe for bronchospasm in patients with a history of asthma.
- Critically unwell patients at risk of cerebral oedema should have NAC infused in a smaller volume of fluid with normal saline, not 5% dextrose. This reduces the risk of potentiating further cerebral oedema. Consult with Toxicology in such cases.
- NAC dose should be increased in consultation with Toxicology or Hepatology when patients are on haemodialysis as NAC is partially cleared during dialysis..

6.0 Drug Interactions

- Nil known

7.0 Dose

N-Acetylcysteine (NAC) is delivered as an intravenous infusion over 20 hours divided into two bags that follow immediately from one another and have different concentrations.

Infusion 1: 200mg/kg NAC in 500ml of 5% dextrose given IV over 4 hours.

Infusion 2: 100mg/kg NAC in 1000ml 5% dextrose given IV over 16 hours.

See appendix 1 for the dose calculator, for any queries contact Clinical Toxicology via switch for advice.

8.0 Preparation

- Add required volume of NAC 200mg/ml to 5% dextrose bag (500ml for first infusion, 1000ml for second infusion.)
    - NAC can be safely administered in normal saline if there are concerns regarding the use of 5% dextrose - for example in hyponatraemia, traumatic brain injury and cerebral oedema.
- Invert bag 10 times to mix solution thoroughly prior to infusing.
- Note: Once an acetylcysteine vial has been punctured the contents may turn from colourless to slightly pink or purple. This does not affect drug potency.

9.0 Administration

- Resuscitation equipment should be accessible at all times.
- Standard infusion pump and IV giving set required.
- Administer via infusion pump at the rates outlined above.

10.0 Monitoring

Observations should be performed before initial infusion and every 15 minutes for the first hour.

After the first hour, and once any reaction has settled, observations can revert to routine practice.

For patients presenting after a single ingestion of immediate-release paracetamol,, where NAC is started,, check an ALT level 2 hrs prior to completion of the second infusion. If the ALT is > 50 U/L (or increasing if baseline ALT is> 50 U/L), continue a further 16-hour infusion of NAC and repeat the ALT level to ensure that the ALT level is static or improving.

If the initial paracetamol level is more than twice the nomogram line, or if modified-release preparation paracetamol has been ingested, a repeat paracetamol level (in addition to the ALT level detailed above) is performed 2 hours prior to completion of the second infusion. If the ALT is > 50 U/L OR if the paracetamol level is > 10 mg/L, a further 16-hour infusion is continued as detailed above.

If the patient has symptoms or signs of hepatotoxicity at the end of the NAC infusion it is recommended to seek Toxicology or Hepatology advice regarding further management.

11.0 Adverse Drug Reactions

Nausea & vomiting

- Common
- Due to NAC or paracetamol ingestion itself.
- Give standard antiemetic therapy.

Non-allergic anaphylactoid reactions (NAAR)

- Modified 2 bag protocol results in less incidence of NAAR compared with the 3-bag protocol.[2]
- May occur during first bag indicated by the following symptoms:
    - Flushing, urticarial rash, itch, tachycardia.
    - Bronchospasm, angioedema, dyspnoea, hypotension, shock (rare). Respiratory symptoms more common in asthmatics.
- Management:
    - Stop NAC infusion temporarily and seek medical review.
    - Antihistamines and bronchodilators may reduce reaction severity.
    - Once symptoms settle recommence infusion at ½ rate and increase to normal rate after 30 minutes.

12.0 Compatibility

- Refer to the Australian Injectable Drugs Handbook for further information on compatibilities and incompatibilities

13.0 Pregnancy and Breastfeeding

- Considered safe to use in pregnancy and breastfeeding
- Contact the Obstetric Medicines Information Service, phone: 6458 2723

Related Documents

Legislation:

- Medicines and Poisons Act 2014
- Medicines and Poisons Regulations 2016
- Health Services Act 2016

Authorising policy and standards

- SCGOPHCG Medicines Management Policy and Procedure
- SCGOPHCG High Risk Medication Policy
- Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy
- OD 0647/16 National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines 2016
- Australian Commission on Safety and Quality in Healthcare. Recommendation for terminology, abbreviations and symbols used in medicines documentation, December 2016
- NSQHS Standards: 1,2,4,5,8

References

1. Chiew AL, et al. Updated guidelines for the management of paracetamol poisoning in Australia and New Zealand. MJA. 2020 Mar 2; 212: 175-183.
2. McNulty R, et al. Fewer adverse effects with a modified two-bag acetylcysteine protocol in paracetamol overdose. Clinical Toxicology. 2018; 18: 618-621.
3. Wong A, et al. Simplification of the standard three-bag intravenous acetylcysteine regimen for paracetamol poisoning results in a lower incidence of adverse drug reactions. Clinical Toxicology. 2016;54:115-119.
4. Chiew AL, et al. Massive paracetamol overdose: an observational study of the effect of activated charcoal and increased acetylcysteine dose (ATOM-2). Clinical Toxicology. 2017;55:1055-1065.
5. Acetylcysteine. Seventh ed. Collingwood 3066, Australia: The Society of Hospital Pharmacists of Australia; 2019
6. The Royal Women's Hospital. Acetylcysteine. In: Pregnancy and Breastfeeding Medicines Guide [Internet]. Parkville (Victoria): The Royal Women's Hospital; 2018 [cited 2023 Aug 11]. Available from: https://thewomenspbmg.org.au/

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Standards Applicable	Insert relevant NSQHS Standards (Version 2):):

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.

Appendix 1 - Dosage Calculator

* NAC dose is calculated on measured body weight rounded up to the nearest 10kg.

* Patients weighing over 110kg should be dosed based on a bodyweight of 110kg.

	Infusion 1 **200mg/kg NAC Added to** **500ml 5% Dextrose** **Given over 4 hours**		**Infusion 2** **100mg/kg NAC Added to** **1000ml 5% Dextrose** **Given over 16 hours**	
Patient's weight (kg)*	**Dose of NAC** (grams)	**Volume of NAC** (ml of 200mg/ml NAC solution)	**Dose of NAC** (grams)	**Volume of NAC** (ml of 200mg/ml NAC solution)
40 kg	8g	40 ml	4g	20 ml
50 kg	10g	50 ml	5g	25 ml
60 kg	12g	60 ml	6g	30 ml
70 kg	14g	70 ml	7g	35 ml
80 kg	16g	80 ml	8g	40 ml
90 kg	18g	90 ml	9g	45 ml
100 kg	20g	100 ml	10g	50 ml
110 kg	22g	110 ml	11g	55ml